

Extensions and Variants

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Grad IO

BLP Extensions: Demographics (Nevo 2000)

- ▶ It is helpful to allow for interactions with consumer demographics (such as income).
- ▶ A few ways to do this:
 - You could just use cross sectional variation in s_{jt} and $\bar{y}_t$ (mean or median income).
 - Better: Divide up your data into additional “markets” by demographics: do you observe s_{jt} at this level? [May not be possible!]
 - Better: Draw y_{it} from a geographic specific income distribution. Draw ν_i from a general distribution of unobserved heterogeneity.
- ▶ Ex: Nevo (2000) Cereal demand sampled individual level y_{it} from geographic specific CPS data
- ▶ Joint distribution of income, income-squared, age, child at home.

$$\beta_i = \bar{\beta} + \Pi y_i + \Sigma \nu_i$$

- ▶ with enough observations on the same product it is possible to include fixed effects

$$\delta_{jt}(\tilde{\theta}_2) = x_{jt}\beta - \alpha p_{jt} + \underbrace{\xi_{jt}}_{\xi_j + \xi_t + \Delta\xi_{jt}}$$

- ▶ What does ξ_j mean in this context?
- ▶ What would ξ_t mean in this context?
- ▶ $\Delta\xi_{jt}$ is now the structural error term, this changes our identification strategy a little.
- ▶ We need instruments that change **within product and across market**.
 - ie: $z_{jt} - \bar{z}_{\cdot t} - \bar{z}_{j\cdot} = \Delta z_{jt}$ has to have some variation left!

Suppose we had additional data on behavior of individuals (in addition to aggregate market shares).

► Examples:

- For some customers have answer to “Which car would you have purchased if the car you bought was not available?”
- Demographic data on purchasers of a single brand.
- Full individual demographic and choice data.

Extensions: Micro Data: Nielsen Panelists

Nielsen data surveys panelists on everything they buy with a UPC code including what store they purchased from.

- ▶ Also tracks household characteristics (Race, Income, Education, HH Size, etc.)
- ▶ Can calculate covariance of characteristics (such as price) with demographics (income, education, etc.) **conditional on purchase**
- ▶ Can calculate purchase probability conditional on demographics: Did you buy any yogurt this trip, week, month, year?

Should we use these as individual data? Or Aggregate data from scanner data with additional moments?

Extensions: Micro Data (Petrin 2002), (microBLP 2004)

- Previously we had moment conditions from orthogonality of structural error (ξ) and (X, Z) in order to form our GMM objective.

$$\mathbb{E}[\xi_{jt}|z_{jt}] = 0 \rightarrow \mathbb{E}[\xi'_{jt}Z_{jt}] = 0$$

- We can incorporate additional information using “micro-moments” or additional moment conditions to match the micro data.
 - $Pr(i \text{ buys } j | y_i \in [0, \$20K]) = c_1$ or $Cov(d_i, s_{ijt}) = c_2$
 - Construct an additional error term ζ_1, ζ_2 and interact that with instruments to form additional moment conditions.
 - Econometrics get tricky when we have a different number of observations for $\mathbb{E}[\zeta' Z_m] = 0$ and $\mathbb{E}[\xi' Z_d] = 0$.
 - May not be able to get covariance of moments taken over different sets of observations!
 - People often assume optimal weight matrices are block diagonal.

Extensions: Complete Micro Data (Grieco, Murry, Pinkse, Sagl 2022)

$$(\hat{\beta}, \hat{\theta}_2, \hat{\delta}) = \arg \min_{\beta, \theta_2, \delta} \underbrace{(-\log \hat{L}(\theta_2, \delta) + \hat{\Pi}(\beta, \delta))}_{\hat{\Omega}(\beta, \theta_2, \delta)}$$

- ▶ $\log \hat{L}(\theta_2, \delta)$ is individual log-likelihood where δ_{jtm} are free parameters and θ_2 are nonlinear parameters.
- ▶ $\hat{\Pi}(\beta, \delta)$ is derived from the moments: $\mathbb{E}[(\delta_{jtm} - \beta x_{jtm} - \alpha p_{jtm}) z_{jtm}] = 0$
- ▶ We don't impose $s_{jt} = \mathfrak{s}_{jt}$

Efficiency requires correcting micro-data to avoid double-counting:

$$\log \hat{L}(\theta, \delta) = \underbrace{\sum_{m=1}^M \sum_{j=0}^{J_m} \sum_{i=1}^{N_m} w_{im} d_{ijm} \log s_{jm}(y_{im})}_{\text{micro}} + \underbrace{\sum_{m=1}^M \sum_{j=0}^{J_m} \left(N_m s_{jm} - \sum_{i=1}^{N_m} w_{im} d_{ijm} \right) \log s_{jm}}_{\text{macro}}$$

Extensions: Second Choices (Conlon, Mortimer, Sarkis 2022)

We need to see at least some $\mathcal{D}_{jk}$

$$\begin{aligned} \min_{s_{ij}, \pi_i} \sum_{(k,j) \in \text{OBS}} (\mathcal{D}_{kj} - D_{kj})^2 + \lambda_1 \cdot \sum_j \left(\mathcal{S}_j - \sum_i \pi_i \cdot s_{ij} \right)^2 \\ \text{subject to } D_{kj} = \sum_{i=1}^I \pi_i \cdot \frac{s_{ij}}{1 - s_{ik}} \cdot \frac{s_{ik}}{s_k} \\ \mathcal{S}_j = \frac{Q_j}{\bar{q}_0 + \sum_{k \in \mathcal{J}_t} Q_k} \\ 0 \leq s_{ij}, \pi_i, s_j, D_{kj} \leq 1, \quad \sum_{i=1}^I \pi_i = 1, \quad \sum_j s_{ij} = 1 \end{aligned}$$

- Semi-parametric (finite-mixture) if we let I grow
- Use cross validation to select # of types I .

Alternative: Vertical Model (Bresnahan 1987)

- Imagine everyone agreed on the quality of the products offered for sale.
- The only thing people disagree on is willingness to pay for quality

$$U_{ij} = \bar{u} + \delta_j - \alpha_i p_j$$

- How do we estimate?

- Sort goods from $p_1 < p_2 < p_3 \dots < p_J$.

It must be that $\delta_1 < \delta_2 < \dots < \delta_J$. Why?

- Normalize o.g. to 0 so that $0 > \delta_1 - \alpha_i p_1$ or $\alpha_i > \delta_1/p_1$.
- $s_0 = F(\infty) - F(\frac{\delta_1}{p_1}) = 1 - F(\frac{\delta_1}{p_1})$ where $F(\cdot)$ is CDF of α_i .
- In general choose j IFF:

$$\frac{\delta_{j+1} - \delta_j}{p_{j+1} - p_j} < \alpha_i < \frac{\delta_j - \delta_{j-1}}{p_j - p_{j-1}}$$
$$s_j = F\left(\frac{\delta_{j+1} - \delta_j}{p_{j+1} - p_j}\right) - F\left(\frac{\delta_j - \delta_{j-1}}{p_j - p_{j-1}}\right)$$

Estimation

- ▶ Choose parameters θ of $F(\cdot)$ in order to best match s_j .
 - Can do MLE $\arg \max_{\theta} \sum_j -s_j \log s_j(\theta)$.
 - Can do least squares $\sum_j (s_j - s_j(\theta))^2$.
 - Can do IV/GMM if I have an instrument for price. $\delta_j = x_j\beta + \xi_j$.
 - Extremely easy when $F \sim \exp(\lambda)$.
- ▶ What about elasticities?
 - When I change the price of j it can only affect (s_{j-1}, s_j, s_{j+1}) .
 - We have set all of the other cross-price elasticities to be zero.
 - If a luxury car and a truck have similar prices, this can create strange substitution patterns.

Pure Characteristics Model: Berry Pakes (2001/2007)

$$u_{ij} = \delta_j + \beta_i x_{jt} + \xi_{jt} + \underbrace{\sigma_\epsilon}_{\rightarrow 0} \cdot \epsilon_{ijt}$$

- ▶ Can think of this like random coefficients model where we take the variance of ϵ to zero.
- ▶ Can think of this a vertical model, with vertical tastes over several characteristics.
 - PCs: everyone prefers more Mhz, more RAM, and more storage but differ in WTP.
 - Possible that there is no PC specific ϵ .
- ▶ Advantages
 - Logit error means there is always some substitution to all other goods.
 - Reality may be you only compete with a small number of competitors.
 - Allows for **crowding** in the product space.
- ▶ Disadvantage: no closed form for s_j , so estimation is extremely difficult.
- ▶ Minjae Song (Homotopy) and Che-Lin Su (MPCC) have made progress using two different approaches.

Even More Flexibility (Fox, Kim, Ryan, Bajari)

Suppose we wanted to nonparametrically estimate $f(\beta_i|\theta)$ instead of assuming that it is normal or log-normal.

$$s_{ij} = \int \frac{\exp[x_j\beta_i]}{1 + \sum_k \exp[x_k\beta_i]} f(\beta_i|\theta)$$

- ▶ Choose a distribution $g(\beta_i)$ that is more spread out than $f(\beta_i|\theta)$
- ▶ Draw several β_s from that distribution (maybe 500-1000).
- ▶ Compute $\hat{s}_{ij}(\beta_s)$ for each draw of β_s and each j .
- ▶ Holding $\hat{s}_{ij}(\beta_s)$ fixed, look for w_s that solve

$$\min_w \left(s_j - \sum_{s=1}^{ns} w_s \hat{s}_{ij}(\beta_s) \right)^2 \quad \text{s.t.} \quad \sum_{s=1}^{ns} w_s = 1, \quad w_s \geq 0 \quad \forall s$$

Even More Flexibility (Fox, Kim, Ryan, Bajari)

- ▶ Like other semi-/non- parametric estimators, when it works it is both flexible and very easy.
- ▶ We are solving a least squares problem with constraints: positive coefficients, coefficients sum to 1.
- ▶ It tends to produce **sparse models** with only a small number of β_s getting positive weights.
 - Why? There is an L_1 penalty term (We are doing **non negative LASSO!**)
- ▶ This is way easier than solving a random coefficients logit model with all but the simplest distributions.
- ▶ There is a bias-variance tradeoff in choosing $g(\beta_i)$.
- ▶ Incorporating parameters that are not random coefficients loses some of the simplicity.
- ▶ I have no idea how to do this with large numbers of fixed effects.

Takes identification arguments in Berry Haile (2014) to the data. Looks at a sieve approximation to

$$\sigma_j^{-1}(\mathcal{S}_t, \tilde{\theta}_2)$$

Using the **Bernstein Polynomials**

- ▶ Bernstein polynomials make it possible to enforce shape restrictions and **monotonicity** which is important
- ▶ Estimates demand for strawberries (organic vs. non-organic)
- ▶ Suggests that both for markups and merger effects we don't have sufficiently flexible demand models.